

ERASMUS UNIVERSITY ROTTERDAM

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Master's Thesis Econometrics and Management Science - Business Analytics and Quantitative
Marketing

Median-of-Means Two-Stage Least Squares Regression

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1 Introduction

2 Median-of-Means (MoM) Estimator

The following sections are adapted from Devroye, Lerasle, Lugosi and Oliveira (2016) and Lugosi and Mendelson (2019)

2.1 Sub-Gaussian Estimators

Definition 1. *Sub-Gaussian Estimators* Let $(X_i)_{i=1}^n$ be a random sample with $\mathbb{E}[X_i] = \mu$ and $\text{Var}[X_i] = \sigma^2$, and let $\bar{\mu}(X)$ denote an estimator for μ . For some constant $L > 0$, $\bar{\mu}(X)$ is said to be a sub-Gaussian estimator for μ if

$$\mathbb{P}\left(|\bar{\mu}(X) - \mu| > L\sigma\sqrt{\frac{\ln(1/\delta)}{n}}\right) \leq \delta \quad \text{for all } \delta \in (0, 1). \quad (1)$$

2.2 The Empirical Mean Estimator

Let $(X_i)_{i=1}^n$ be a random sample with $\mathbb{E}[X_i] = \mu$ and $\text{Var}[X_i] = \sigma^2 < \infty$. The empirical mean is defined as

$$\bar{\mu} = \frac{1}{n} \sum_{i=1}^n X_i.$$

By the Central Limit Theorem,

$$\sqrt{n}(\bar{\mu} - \mu) \xrightarrow{d} \mathcal{N}(0, \sigma^2),$$

which implies that for any fixed $\delta \in (0, 1)$,

$$\mathbb{P}\left(|\bar{\mu} - \mu| > \frac{\sigma\Phi^{-1}(1 - \delta/2)}{\sqrt{n}}\right) \rightarrow \delta \quad \text{as } n \rightarrow \infty.$$

Using the bound $\Phi^{-1}(1 - \delta/2) \leq \sqrt{2\ln(2/\delta)}$, it follows that

$$\lim_{n \rightarrow \infty} \mathbb{P}\left(|\bar{\mu} - \mu| > \sigma\sqrt{\frac{2\ln(2/\delta)}{n}}\right) \leq \delta,$$

so $\bar{\mu}$ is an asymptotically sub-Gaussian estimator in the sense of (1) with constant $L = \sqrt{2}$.

While $\bar{\mu}$ is asymptotically sub-Gaussian, it is not a sub-Gaussian estimator for finite samples; Chebyshev's inequality gives

$$\mathbb{P}\left(|\bar{\mu} - \mu| > \sigma\sqrt{\frac{1}{n\delta}}\right) \leq \delta$$

shows that the dependence on δ is polynomial rather than logarithmic, which is substantially worse than (1) for small δ . As such, the empirical mean is insufficient as a sub-Gaussian estimator

for random variables with only finite first and second moments.

2.3 The Median-of-Means Estimator

The median-of-means (MoM) estimator is not new, being proposed in various forms, first by Nemirovskij and Yudin (1983) and Jerrum, Valiant and Vazirani (1986). The algorithm works as follows

Algorithm 1: Median-of-Means Estimator

Input: Sample $(X_i)_{i=1}^n$, number of blocks k

Output: Estimate $\hat{\mu}$

Partition $(X_i)_{i=1}^n$ into k blocks $B_1, \dots, B_k$ of size $m = \lfloor n/k \rfloor$;

for $j = 1, \dots, k$ **do**

 Compute the block mean $\bar{\mu}_j = \frac{1}{m} \sum_{i \in B_j} X_i$;

return $\hat{\mu} = \text{median}(\bar{\mu}_1, \dots, \bar{\mu}_k)$;

The MoM estimator satisfies

$$\mathbb{P}\left(|\hat{\mu} - \mu| > \sigma \sqrt{\frac{4}{m}}\right) \leq e^{-k/8},$$

and setting $k = \lceil 8 \ln(1/\delta) \rceil$, this becomes

$$\mathbb{P}\left(|\hat{\mu} - \mu| > \sigma \sqrt{\frac{32 \ln(1/\delta)}{n}}\right) \leq \delta \quad \text{for all } \delta \in (0, 1) \quad (2)$$

which is sub-Gaussian with $L = \sqrt{32}$

3 Median-of-Means Two-Stage Least Squares (MoM-2SLS)

We first explore the MoM estimator in a univariate setting, with one instrument and one variable.

3.1 Univariate MoM-2SLS

In the univariate setting, the 2SLS estimator collapses to the Wald/IV estimator

$$\hat{\beta}_{IV} = \frac{\sum_{i=1}^n Z_i Y_i}{\sum_{i=1}^n Z_i X_i} = \frac{\bar{S}_{ZY}}{\bar{S}_{ZX}}.$$

This is a ratio of two sample means. To solve this, we can utilise two MoM-type estimators. The first replaces each sample mean with its MoM counterpart; we call this the ratio-of-medians estimator. The second returns the median of the Wald/IV estimate of each block, we call this the median-of-ratios estimator.

3.2 Setup

Let $(Y_i, X_i, Z_i)_{i=1}^n$ be an i.i.d. random sample satisfying the structural equation

$$Y_i = \beta X_i + \varepsilon_i,$$

where $\mathbb{E}[Z_i \varepsilon_i] = 0$ (exogeneity) and $\mu_{ZX} := \mathbb{E}[Z_i X_i] \neq 0$ (relevance). Define

$$\mu_{ZY} := \mathbb{E}[Z_i Y_i], \quad \sigma_{ZY}^2 := \text{Var}(Z_i Y_i), \quad \sigma_{ZX}^2 := \text{Var}(Z_i X_i), \quad \sigma_{Z\varepsilon}^2 := \text{Var}(Z_i \varepsilon_i),$$

all assumed finite. Both estimators partition the sample into k blocks $B_1, \dots, B_k$ of size $m = \lfloor n/k \rfloor$ and compute block means

$$\hat{S}_{ZY}^{(j)} = \frac{1}{m} \sum_{i \in B_j} Z_i Y_i, \quad \hat{S}_{ZX}^{(j)} = \frac{1}{m} \sum_{i \in B_j} Z_i X_i.$$

They differ only in how the block-level quantities are aggregated.

3.3 The Two Estimators

Definition 2 (Ratio-of-Medians (RoM)).

$$\hat{\beta}_{\text{RoM}} = \frac{\tilde{S}_{ZY}}{\tilde{S}_{ZX}}, \quad \text{where } \tilde{S}_{ZY} = \text{median}_j(\hat{S}_{ZY}^{(j)}), \quad \tilde{S}_{ZX} = \text{median}_j(\hat{S}_{ZX}^{(j)}).$$

The medians are taken separately and the ratio is formed afterwards.

Definition 3 (Median-of-Ratios (MoR)).

$$\hat{\beta}_{\text{MoR}} = \text{median}_j \left(\hat{\beta}^{(j)} \right), \quad \text{where } \hat{\beta}^{(j)} = \frac{\hat{S}_{ZY}^{(j)}}{\hat{S}_{ZX}^{(j)}}.$$

The ratio is formed within each block and the median is taken over the block-level estimates.

The proofs for these results can be found in the tablet notes.

3.4 Comparison of the Two Estimators

3.4.1 Effective Variance

The RoM bound has $\sigma_{ZY} + |\beta|\sigma_{ZX}$ in the numerator while the MoR has $\sigma_{Z\varepsilon}$. Using $Z_i Y_i = \beta Z_i X_i + Z_i \varepsilon_i$, it can be shown that

$$\sigma_{Z\varepsilon} \leq \sigma_{ZY} + |\beta|\sigma_{ZX}.$$

Thus, the MoR bound is tighter. The proof is found in the tablet notes.

3.4.2 Number of Blocks

The RoM requires $k = \lceil 8 \ln(2/\delta) \rceil$ while the MoR requires $k = \lceil 8 \ln(1/\delta) \rceil$. The difference is the factor $\ln(2/\delta)$ versus $\ln(1/\delta)$, which amounts to an additive constant $8 \ln 2 \approx 5.5$ in k .

3.4.3 Instrument Strength Condition

The most interesting result is the existence of an instrument strength condition for the guarantee of sub-Gaussian bounds.

	Ratio-of-Medians	Median-of-Ratios
Condition on m	$m > 4\sigma_{ZX}^2/\mu_{ZX}^2$	$m \geq 32\sigma_{ZX}^2/\mu_{ZX}^2$
Minimum n	$\frac{32\sigma_{ZX}^2}{\mu_{ZX}^2} \ln(2/\delta)$	$\frac{256\sigma_{ZX}^2}{\mu_{ZX}^2} \ln(1/\delta)$

The MoR condition is eight times stronger

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